Apply filters to SQL queries

Project description

In this project, I worked as a security analyst in a large organization. My task was to investigate possible security issues related to login attempts and employee computers. I used SQL queries to find important information in the log_in_attempts and employees tables. I used different SQL filters to help identify security problems and employees who needed computer updates.

Retrieve after hours failed login attempts

My team discovered possible security issues related to failed login attempts after business hours. I needed to find all failed login attempts after 18:00.

I used the following SQL query:

```
MariaDB [organization]> SELECT *
    -> FROM log_in_attempts
    -> WHERE login_time > '18:00' AND success = FALSE;
+----------+----------+------------+------------+---------+----------------+---------+
| event_id | username | login_date | login_time | country | ip_address     | success |
+----------+----------+------------+------------+---------+----------------+---------+
|        2 | apatel   | 2022-05-10 | 20:27:27   | CAN     | 192.168.205.12 |       0 |
|       18 | pwashing | 2022-05-11 | 19:28:50   | US      | 192.168.66.142 |       0 |
|       20 | tshah    | 2022-05-12 | 18:56:36   | MEXICO  | 192.168.109.50 |       0 |
```

This query shows all failed login attempts after 18:00. First, I used SELECT * to get all information from the log_in_attempts table. Then, I used WHERE with AND to check two conditions. The first condition, login_time > '18:00', finds logins after 18:00. The second condition, success = FALSE, finds unsuccessful login attempts.

The query returned **19 failed login attempts**.

Retrieve login attempts on specific dates

A suspicious event happened on May 9, 2022. My task was to check all login attempts from that day and the day before.

I used the following SQL query:

```
MariaDB [organization]> SELECT *
    -> FROM log_in_attempts
    -> WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';
+----------+----------+------------+------------+---------+-----------------+---------+
| event_id | username | login_date | login_time | country | ip_address      | success |
+----------+----------+------------+------------+---------+-----------------+---------+
|        1 | jrafael  | 2022-05-09 | 04:56:27   | CAN     | 192.168.243.140 |       0 |
|        3 | dkot     | 2022-05-09 | 06:47:41   | USA     | 192.168.151.162 |       0 |
|        4 | dkot     | 2022-05-08 | 02:00:39   | USA     | 192.168.178.71  |       0 |
```

This query finds login attempts from May 8 and May 9, 2022. I used SELECT * to get all records from the log_in_attempts table. Then, I used WHERE with OR to find logins from either date. The OR operator allows the query to return results that match at least one of the two conditions. The query returned 75 login attempts.

Retrieve login attempts outside of Mexico

My team found suspicious login activity that did not come from Mexico. I needed to find all login attempts from other countries.

I used the following SQL query:

```
MariaDB [organization]> SELECT *
    -> FROM log_in_attempts
    -> WHERE NOT country LIKE 'MEX%';
+----------+----------+------------+------------+---------+-----------------+---------+
| event_id | username | login_date | login_time | country | ip_address      | success |
+----------+----------+------------+------------+---------+-----------------+---------+
|        1 | jrafael  | 2022-05-09 | 04:56:27   | CAN     | 192.168.243.140 |       0 |
|        2 | apatel   | 2022-05-10 | 20:27:27   | CAN     | 192.168.205.12  |       0 |
|        3 | dkot     | 2022-05-09 | 06:47:41   | USA     | 192.168.151.162 |       0 |
```

This query finds login attempts from countries other than Mexico. I used NOT to exclude Mexico from the results. I also used LIKE with the pattern 'MEX%' because Mexico can be written as MEX or MEXICO in the database. The % symbol matches any number of characters after MEX. The query returned 144 login attempts.

Retrieve employees in Marketing

My team needed to update computers for employees in the Marketing department who worked in the East building. I had to find information about these employees.

I used the following SQL query:

```
MariaDB [organization]> SELECT *
    -> FROM employees
    -> WHERE department = 'Marketing' AND office LIKE 'East%';
+-------------+-------------+----------+------------+----------+
| employee_id | device_id   | username | department | office   |
+-------------+-------------+----------+------------+----------+
|        1000 | a320b137c219 | elarson  | Marketing  | East-170 |
|        1052 | a192b174c940 | jdarosa  | Marketing  | East-195 |
|        1075 | x573y883z772 | fbautist | Marketing  | East-267 |
```

This query finds employees from the Marketing department in the East building. I used SELECT * to get information from the employees table. Then, I used AND to combine two conditions. The first condition finds employees from Marketing, and the second uses LIKE 'East%' to find all offices in the East building. The first employee's username was elarson.

Retrieve employees in Finance or Sales

My team needed to perform another security update on computers in the Finance and Sales departments. I had to find all employees from these departments.

I used the following SQL query:

```
MariaDB [organization]> SELECT *
    -> FROM employees
    -> WHERE department = 'Finance' OR department = 'Sales';
+-------------+-------------+----------+------------+-------------+
| employee_id | device_id   | username | department | office      |
+-------------+-------------+----------+------------+-------------+
|        1003 | d394e816f943 | sgilmore | Finance    | South-153   |
|        1007 | h174i497j413 | wjaffrey | Finance    | North-406   |
|        1008 | i858j583k571 | abernard | Finance    | South-170   |
```

This query finds employees who work in Finance or Sales. I used SELECT * to get all information from the employees table. Then, I used WHERE with OR because I wanted to include employees from either department. The first condition checks Finance, and the second checks Sales. The first employee from Sales had the username lrodriqu.

Retrieve all employees not in IT

My team needed to make one more security update. Employees in the Information Technology department already had this update, so I needed to find employees from other departments.

I used the following SQL query:

```
MariaDB [organization]> SELECT *
    -> FROM employees
    -> WHERE NOT department = 'Information Technology';
+-------------+--------------+----------+-----------------+-------------+
| employee_id | device_id    | username | department      | office      |
+-------------+--------------+----------+-----------------+-------------+
|        1000 | a320b137c219 | elarson  | Marketing       | East-170    |
|        1001 | b239c825d303 | bmoreno  | Marketing       | Central-276 |
|        1002 | c116d593e558 | tshah    | Human Resources | North-434   |
```

This query finds all employees who do not work in the Information Technology department. First, I selected all data from the employees table. Then, I used WHERE with NOT to exclude employees from the IT department. The query returned 161 employees.

Summary

In this project, I used SQL queries to investigate possible security issues and find information about employee computers. I worked with two tables: log_in_attempts and employees. I used AND, OR, and NOT to filter data based on different conditions. I also used LIKE with the % wildcard to search for specific patterns. These SQL skills can help security analysts investigate suspicious activity and improve system security.